

Trial Task: Analysing Student–LLM Collaboration Quality

Context

Large Language Models (LLMs) are changing how students learn. As their influence grows, Higher Education Institutions are working to prepare future professionals with the competencies to engage in a collaborative process with the LLM-based tools. However, using an LLM and *collaborating* with one are not the same thing. Within the DFG/ANR project *Analytics for Learning with Machines* (ALMA), the TUM side focuses on developing **analytical methods and computational indicators** to characterise the quality of student–LLM collaboration. The central question is: given a trace of student–LLM interaction, what can we measure, model, or infer collaboration quality in a student-LLM dyad in a learning scenario?

Objective of the Task

The goal of this task is to analyse a set of student–LLM interactions and propose indicators that could characterise the quality of the collaboration. With your analysis, we want to see:

- How you conceptualize “collaboration quality” between a student and an LLM
- How you operationalize that concept into observable or computable indicators
- How you reason about what the data can and cannot tell us

The task is divided into 2 parts. See descriptions below.

Task Description

Data

You are provided with two datasets from the same cohort of 30 students enrolled in a higher education course. The students were given a writing task: to produce a short synthesis (300–350 words) of the two texts they were given to read, the texts were about predictive modelling in education. Students had access to an LLM chatbot and were free to use it however they wished during the task.

Dataset 1 – Conversations (alma_conversations.csv): Interaction logs between each student and the LLM during the task. Each row represents one exchange (student request + LLM response). Fields: student_id, student_request, llm_response, timestamp (each student had 0-60 minutes roughly to complete the task).

Dataset 2 – Essays (alma_essays.csv): The final written product submitted by each student after the task. Fields: student_id, essay.

Part 1 – Data Analysis

Explore the data and propose indicators that could characterise the quality of the student–LLM collaboration. You have total freedom in how you approach this. Your analysis should **propose indicators** that could characterise collaboration quality, apply the indicators to the data and present your findings. You have total freedom in your choice of programming language and tools.

Part 2 – Report: Conceptualisation, Reflection, and Analysis (3 pages max)

1. Define your view of collaboration quality

- What does “high-quality collaboration” between a student and an LLM in a learning task look like, in your view?

- What distinguishes effective use of an LLM from ineffective use, from a learning perspective?
- You may draw on theories or frameworks from collaborative learning, dialogue research, human–AI interaction, or other relevant fields.

2. Justify your indicators

- Why did you choose these specific indicators?
- How do they relate to your conceptualisation of collaboration quality?
- How are your indicators describing a learner-LLM system? What are their strengths and limitations?

3. Reflect on findings and future directions

- What did your analysis reveal? Were there patterns, surprises, or ambiguities?
- How would you improve, extend or change your analysis?
- What data would you need to evaluate collaboration quality at scale?

Time Expectations

This task is designed to take no more than 2 working days, we will give you 5 working days to work on it.

Evaluation Criteria

Overall, we will value clarity of thinking and analytical reasoning more than technical sophistication. Candidates will be evaluated based on:

- Understanding of collaborative learning and human–AI interaction concepts
- Ability to operationalise abstract theoretical concepts into measurable indicators, with particular attention to system-level metrics
- Quality and clarity of your analysis
- Ability to reflect critically on limitations and future work
- Overall coherence between conceptual framing and analytical choices